

AI WeatherWise

AI-Augmented Backend Development Project

PHASE 5 – PROJECT DEVELOPMENT

Team No.	05
Team ID	SWTID-2026-9139
Project Title	AI WeatherWise
Team Leader	HIBA FATHIMA S
Team Members	DIVYA SREE V, LINCYA A H, RESHMA J

1. Project Development Overview

The Project Development phase focused on implementing AI WeatherWise as a modular backend application using Node.js, Express.js, MongoDB, and JWT authentication. The team converted the approved design into working APIs, integrated external weather services, and developed secure backend functionality.

Development Objective

Build a secure RESTful backend capable of authenticating users, storing data in MongoDB, retrieving live weather information, and generating weather-based recommendations.

2. Development Environment

Component	Technology
Runtime	Node.js
Framework	Express.js
Database	MongoDB Atlas
ODM	Mongoose
Authentication	JWT + bcrypt
Testing	Thunder Client

3. Backend Module Development

Authentication Module

Implemented user registration, secure login, JWT token generation, protected routes, and bcrypt password hashing.

Weather Module

Integrated external Weather APIs to retrieve live weather information including temperature, humidity, wind speed, and weather conditions.

Location Module

Developed MongoDB collections that allow users to save and manage multiple favourite locations.

Recommendation Module

Built a recommendation workflow that generates weather-based suggestions with fallback handling when external services are unavailable.

4. Folder Structure

Project Structure

```
src/  
├── config  
├── controllers  
├── middleware  
├── models  
├── routes  
├── services  
├── app.js  
└── server.js
```

5. Database Implementation

MongoDB Atlas was used as the cloud database for storing user accounts and favourite locations.

Collection	Purpose
User	Stores user details
Location	Stores favourite cities
Authentication	JWT verification
Security	Hashed passwords
Relationship	One User → Many Locations

6. Development Workflow

- Configured Node.js and Express.js.
- Connected MongoDB Atlas.
- Created authentication APIs.
- Developed weather retrieval APIs.
- Implemented favourite location management.
- Integrated recommendation workflow.
- Verified APIs using Thunder Client.
- Uploaded the completed backend to GitHub.

7. Features Successfully Developed

Feature	Status
User Registration	Completed
User Login	Completed
JWT Authentication	Completed
Weather Retrieval	Completed
Favourite Locations	Completed
Recommendation Endpoint	Completed
GitHub Repository	Completed

8. Project Development Outcome

The Project Development phase successfully transformed the planned backend architecture into a functional application. AI WeatherWise now includes secure authentication, MongoDB integration, weather retrieval, recommendation processing, and a modular codebase that supports testing, documentation, and final project demonstration.